

CSE213

MICROCONTROLLER PROGRAMMING

Data Movement Instructions

Introduction

- This chapter concentrates on common data movement instructions.

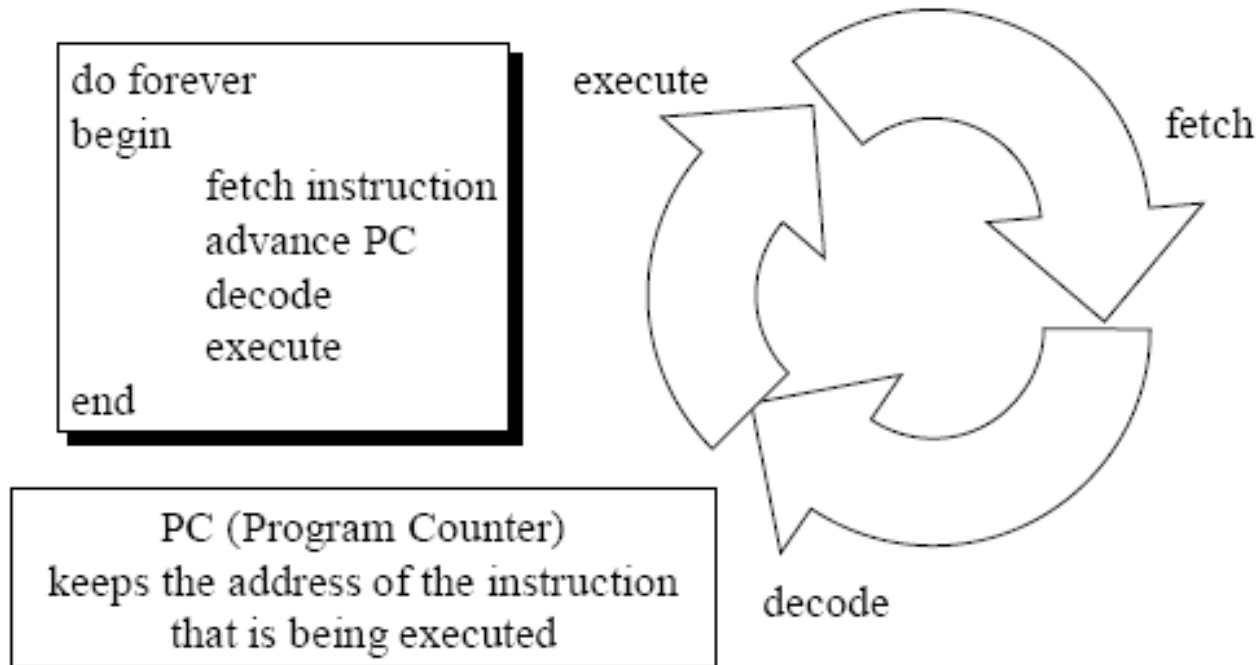
Chapter Objectives

Upon completion of this chapter, you will be able to:

- Explain the operation of each data movement instruction with applicable addressing modes.
- Select the appropriate assembly language instruction to accomplish a specific data movement task.

Instruction Cycle

Basic Instruction Cycle



Machine Language

- Native binary code microprocessor uses as its instructions to control its operation.
 - instructions vary in length from 1 to 13 bytes
- Over 100,000 variations of machine language instructions.
 - there is no complete list of these variations
- Some bits in a machine language instruction are given; remaining bits are determined for each variation of the instruction.

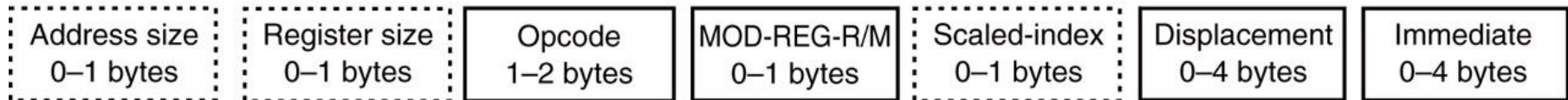
Instruction Format

16-bit instruction mode



(a)

32-bit instruction mode (80386 through Pentium 4 only)



(b)

Figure 4–1 The formats of the 8086–Core2 instructions. (a) The 16-bit form and (b) the 32-bit form.

The Opcode

- Selects the operation (addition, subtraction, etc.,) performed by the microprocessor.
- Figure below illustrates the general form of the first opcode byte of many instructions.
 - first 6 bits of the first byte are the binary opcode
 - remaining 2 bits indicate the **direction** (D) of the data flow, and indicate whether the data are a byte or a word (W)

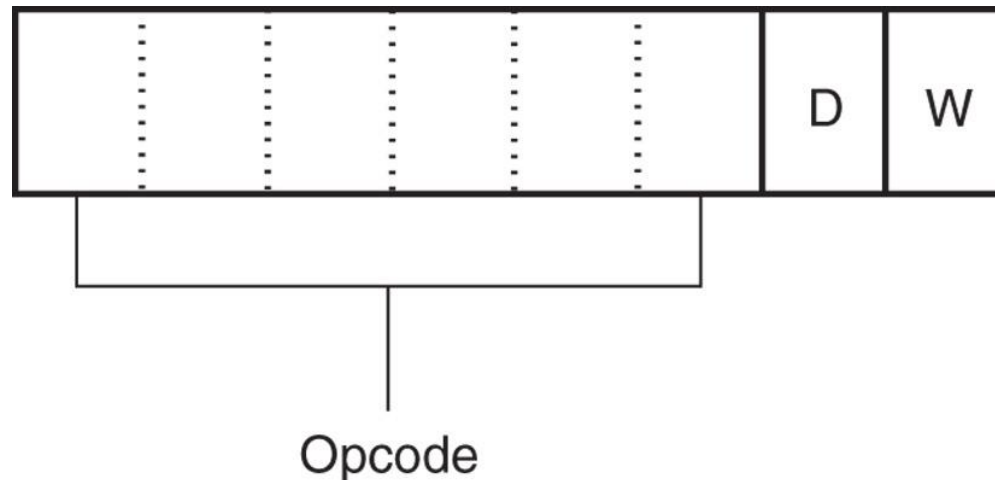


Figure 4–3 Byte 2 of many machine language instructions, showing the position of the MOD, REG, and R/M fields.



MOD Field

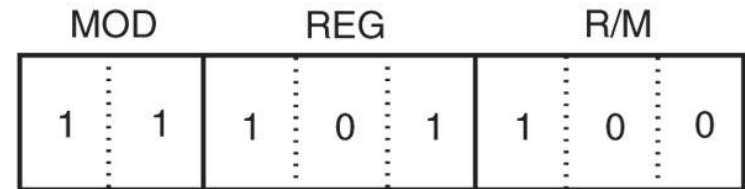
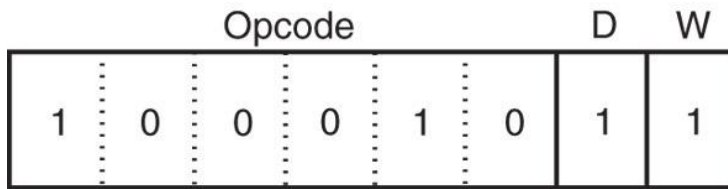
- Specifies addressing mode (MOD) and whether a displacement is present with the selected type.
 - If MOD=11, register-addressing mode is selected
 - Register addressing specifies a register instead of a memory location, using the R/M field
- If the MOD field contains a 00, 01, or 10, the R/M field selects one of the data memory-addressing modes.

- All 8-bit displacements are sign-extended into 16-bit displacements when the processor executes the instruction.
 - if the 8-bit displacement is 00H–7FH (positive), it is sign-extended to 0000H–007FH before adding to the offset address
 - if the 8-bit displacement is 80H–FFH (negative), it is sign-extended to FF80H–FFFFH

Register Assignments

- Suppose a 2-byte instruction, 8BECH, appears in a machine language program.
- In 16-bit mode, this instruction is converted to binary and placed in the instruction format of bytes 1 and 2, as illustrated in Figure 4–4.

Figure 4–4 The 8BEC instruction placed into bytes 1 and 2 formats from Figures 4–2 and 4–3. This instruction is a MOV BP,SP.



Opcode = MOV

D = Transfer to register (REG)

W = Word

MOD = R/M is a register

REG = BP

R/M = SP

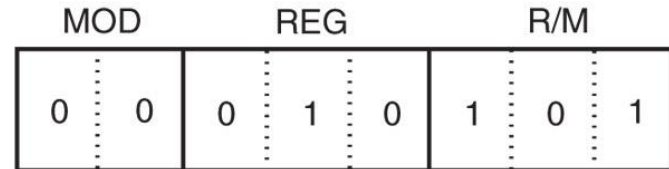
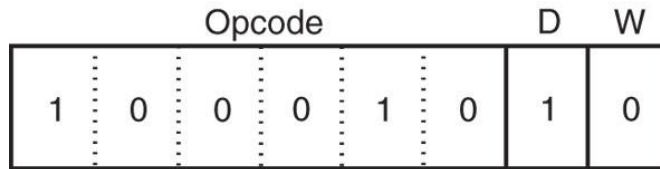
— the opcode is 100010, a MOV instruction

- D and W bits are a logic 1, so a word moves into the destination register specified in the REG field
- REG field contains 101, indicating register BP, so the MOV instruction moves data into register BP

R/M Memory Addressing

- If the MOD field contains a 00, 01, or 10, the R/M field takes on a new meaning.
- Figure 4–5 illustrates the machine language version of the 16-bit instruction MOV DL,[DI] or instruction (8A15H).
- This instruction is 2 bytes long and has an opcode 100010, D=1 (to REG from R/M), W=0 (byte), MOD=00 (no displacement), REG=010 (DL), and R/M=101 ([DI]).

Figure 4–5 A MOV DL,[DI] instruction converted to its machine language form.



Opcode = MOV

D = Transfer to register (REG)

W = Byte

MOD = No displacement

REG = DL

R/M = DS:[DI]

- If the instruction changes to MOV DL, [DI+1], the MOD field changes to 01 for 8-bit displacement
- first 2 bytes of the instruction remain the same
- instruction now becomes 8A5501H instead of 8A15H

- Because the MOD field contains a 11, the R/M field also indicates a register.
- R/M = 100(SP); therefore, this instruction moves data from SP into BP.
 - written in symbolic form as a MOV BP,SP instruction
- The assembler program keeps track of the register- and address-size prefixes and the mode of operation.

Data Types

- In order to tell the assembler about data type, these prefixes should be used:
 - **BYTE PTR** - for byte.
 - **WORD PTR** - for word (two bytes)
- Examples:
 - **MOV AL, BYTE PTR [BX]** ; byte access
 - **MOV CX, WORD PTR [BX]** ; word access
- Assembler supports shorter prefixes as well:
 - **B.** - for **BYTE PTR**
 - **W.** - for **WORD PTR**
- In certain cases the assembler can calculate the data type automatically.

MOV Instruction

- Copies the **second operand** (source) to the **first operand** (destination).
- The source operand can be an immediate value, general-purpose register or memory location.
- The destination register can be a general-purpose register, or memory location.
- Both operands must be the same size, which can be a byte or a word.
- the **MOV** instruction cannot be used to set the value of the **CS** and **IP** registers.

Operands of MOV

- These types of operands are supported:
 - MOV REG, memory
 - MOV memory, REG
 - MOV REG, REG
 - MOV memory, immediate
 - MOV REG, immediate
- **REG:** AX, BX, CX, DX, AH, AL, BL, BH, CH, CL, DH, DL, DI, SI, BP, SP.
- **memory:** [BX], [BX+SI+7], variable, etc.
- **immediate:** 5, -24, 3Fh, 10001101b, etc.

Segment Register Operands

- For segment registers only these types of **MOV** are supported:
 - MOV SREG, memory
 - MOV memory, SREG
 - MOV REG, SREG
 - MOV SREG, REG
- **SREG**: DS, ES, SS, and only as second operand: CS.
- **REG**: AX, BX, CX, DX, AH, AL, BL, BH, CH, CL, DH, DL, DI, SI, BP, SP.
- **memory**: [BX], [BX+SI+7], variable, etc.

MOV Example

ORG 100h	; this directive required
MOV AX, 0B800h	; set AX to hexadecimal value of B800h.
MOV DS, AX	; copy value of AX to DS.
MOV CL, 'A'	; set CL to ASCII code of 'A', it is 41h.
MOV CH, 11011111b	; set CH to binary value.
MOV BX, 15Eh	; set BX to 15Eh.
MOV [BX], CX	; copy contents of CX to memory at B800:015E
RET	; returns to operating system.

Variables

- Syntax for a variable declaration:
 - name **DB** value
 - name **DW** value
- **DB** - stays for Define Byte.
- **DW** - stays for Define Word.
- name - can be any letter or digit combination, though it should start with a letter. It's possible to declare unnamed variables by not specifying the name (this variable will have an address but no name).
- value - can be any numeric value in any supported numbering system (hexadecimal, binary, or decimal), or "?" symbol for variables that are not initialized.

Example

ORG 100h

MOV AL, var1

MOV BX, var2

RET ; stops the program.

var1 DB 7

var2 DW 1234H

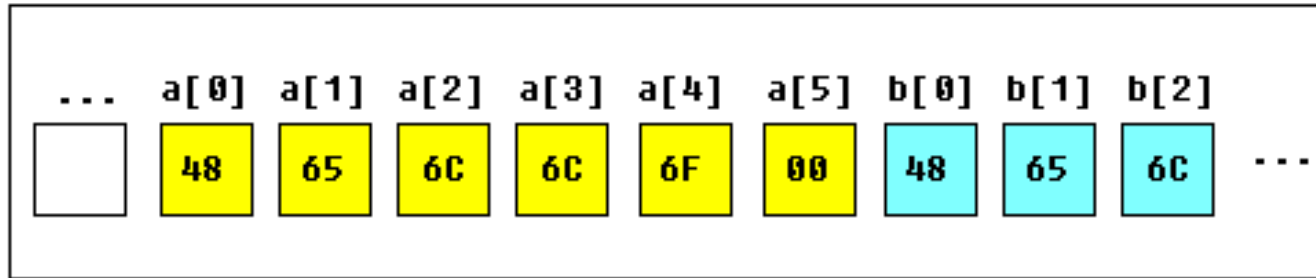
ORG Directive

- **ORG 100h** is an assembler directive (it tells assembler how to handle the source code).
- It tells assembler that the executable file will be loaded at the **offset** of 100h (256 bytes), so assembler should calculate the correct address for all variables when it replaces the variable names with their **offsets**.
- Directives are never converted to any real **machine code**.
- Why executable file is loaded at **offset** of **100h**?
- Operating system keeps some data about the program in the first 256 bytes of the **CS** (code segment), such as command line parameters and etc.

Arrays

- Arrays can be seen as chains of variables.
- A text string is an example of a byte array, each character is represented as an ASCII code value (0..255).
- Examples:
 - **a DB 48h, 65h, 6Ch, 6Ch, 6Fh, 00h**
 - **b DB 'Hello', 0**
- **b** is an exact copy of the **a** array, when assembler sees a string inside quotes it automatically converts it to set of bytes.

Accessing Array Elements



- You can access the value of any element in array using square brackets, for example:
 - MOV AL, a[3]
- You can also use any of the memory index registers **BX, SI, DI, BP**, for example:
 - MOV SI, 3
 - MOV AL, a[SI]

Declaring Large Arrays

- If you need to declare a large array you can use **DUP** operator.
- The syntax for **DUP**:
 - number DUP (value(s))
 - number - number of duplicate to make (any constant value).
 - value - expression that DUP will duplicate.
- Example:
c DB 5 DUP(9)
is an alternative way of declaring:
c DB 9, 9, 9, 9, 9

Declaring Large Arrays

- One more example:
 d DB 5 DUP(1, 2)
 is an alternative way of declaring:
 d DB 1, 2, 1, 2, 1, 2, 1, 2, 1, 2
- Of course, you can use **DW** instead of **DB** if it's required to keep values larger than 255, or smaller than -128.
- **DW** cannot be used to declare strings.

Getting the Address of a Variable

- The **LEA** instruction can be used to get the offset address of a variable.
- Getting the address of the variable can be very useful in some situations, for example when you need to pass parameters to a procedure.

Example

ORG 100h

MOV AL, VAR1 ; check value of VAR1 by moving it to AL.

LEA BX, VAR1 ; get address of VAR1 in BX.

MOV BYTE PTR [BX], 44h ; modify the contents of VAR1.

MOV AL, VAR1 ; check value of VAR1 by moving it to AL.

RET

VAR1 DB 22h

END

Constants

- Constants are just like variables, but they exist only until your program is compiled (assembled).
- After definition of a constant its value cannot be changed.
- To define constants **EQU** directive is used:
name EQU < any expression >
- Example:
k EQU 5
MOV AX, k
- The above example is functionally identical to code:
MOV AX, 5

PUSH

- Always transfers 2 bytes of data to the stack;
 - 80386 and above transfer 2 or 4 bytes
- PUSH instruction copies contents of the internal register set, except the segment registers, to the stack.
- PUSH (push all) instruction copies the registers to the stack in the following order: AX, CX, DX, BX, SP, BP, SI, and DI.

- PUSHF (**push flags**) instruction copies the contents of the flag register to the stack.
- PUSHAD and POPAD instructions push and pop the contents of the 32-bit register set in 80386 - Pentium 4.
 - PUSHA and POPA instructions do not function in the 64-bit mode of operation for the Pentium 4

POP

- Performs the inverse operation of PUSH.
- POP removes data from the stack and places it in a target 16-bit register, segment register, or a 16-bit memory location.
 - not available as an immediate POP
- POPF (pop flags) removes a 16-bit number from the stack and places it in the flag register;
 - POPFD removes a 32-bit number from the stack and places it into the extended flag register

- POPA (pop all) removes 16 bytes of data from the stack and places them into the following registers, in the order shown: DI, SI, BP, SP, BX, DX, CX, and AX.
 - reverse order from placement on the stack by PUSHA instruction, causing the same data to return to the same registers

4-4 STRING DATA TRANSFERS

- Before the string instructions are presented, the operation of the D flag-bit (direction), DI, and SI must be understood as they apply to the string instructions.

The Direction Flag

- The direction flag (D, located in the flag register) selects auto-increment or auto-decrement operation for the DI and SI registers during string operations.
 - used only with the string instructions
- The CLD instruction clears the D flag and the STD instruction sets it .
 - CLD instruction selects the auto-increment mode and STD selects the auto-decrement mode

DI and SI

- During execution of string instruction, memory accesses occur through DI and SI registers.
 - DI offset address accesses data in the extra segment for all string instructions that use it
 - SI offset address accesses data by default in the data segment

LODSB Instruction

- Load byte at DS:[SI] into AL. Update SI.

Algorithm:

$AL = DS:[SI]$

- if $DF = 0$ then
 - $SI = SI + 1$
- else
 - $SI = SI - 1$

Example

ORG 100h

LEA SI, a1

MOV CX, 5

MOV AH, 0Eh

m: LODSB

INT 10h

LOOP m

RET

a1 DB 'H', 'e', 'l', 'l', 'o'

STOSB Instruction

- Store byte in AL into ES:[DI]. Update DI.

Algorithm:

ES:[DI] = AL

- if DF = 0 then
 - DI = DI + 1
- else
 - DI = DI - 1

Example

ORG 100h

LEA DI, a1

MOV AL, 12h

MOV CX, 5

REP

STOSB

RET

a1 DB 5 dup(0)

MOVSb Instruction

- Copy byte at DS:[SI] to ES:[DI]. Update SI and DI.

Algorithm:

ES:[DI] = DS:[SI]

- if DF = 0 then
 - SI = SI + 1
 - DI = DI + 1
- else
 - SI = SI - 1
 - DI = DI - 1

Example

ORG 100h

CLD

LEA SI, a1

LEA DI, a2

MOV CX, 5

REP

MOVSB

RET

a1 DB 1,2,3,4,5

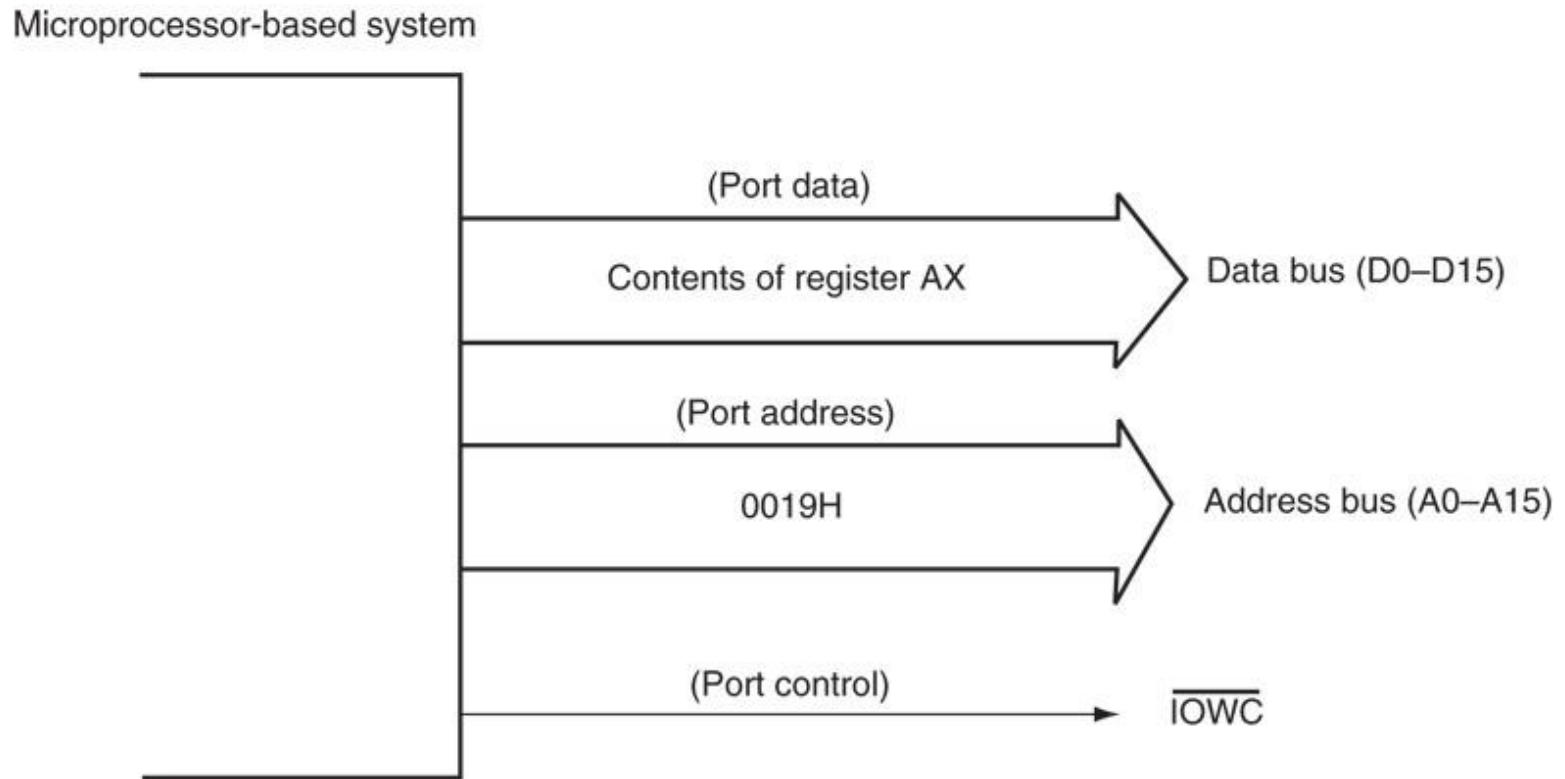
a2 DB 5 DUP(0)

IN and OUT

- IN & OUT instructions perform I/O operations.
- Contents of AL, AX, or EAX are transferred only between I/O device and microprocessor.
 - an IN instruction transfers data from an external I/O device into AL, AX, or EAX
 - an OUT transfers data from AL, AX, or EAX to an external I/O device

- Two forms of I/O device (port) addressing:
- *Fixed-port addressing* allows data transfer between AL, AX, or EAX using an 8-bit I/O port address.
 - port number follows the instruction's opcode
- *Variable-port addressing* allows data transfers between AL, AX, or EAX and a 16-bit port address.
 - the I/O port number is stored in register DX, which can be changed (varied) during the execution of a program.

Figure 4–20 The signals found in the microprocessor-based system for an OUT 19H,AX instruction.





The Intel Microprocessors: 8086/8088, 80186/80188, 80286, 80386, 80486 Pentium, Pentium Pro Processor, Pentium II, Pentium, 4, and Core2 with 64-bit Extensions Architecture, Programming, and Interfacing, Eighth Edition
Barry B. Brey

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